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MINI-REVIEW ARTICLE

Impact of Microgravity on Virulence, Antibiotic Resistance and Gene Expression in Beneficial and Pathogenic Microorganisms

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Abstract: During space missions, the impact of the space conditions (both microgravity and radiation) on physiologic and metabolic aspects of the microbiota of astronauts' bodies should be considered. Changes depend on the mission's duration, types of organisms, and ecology. Reported alterations are related to changes in morphology, growth, gene expression, and physiology of cells, resulting in increased virulence, acid, antibiotic resistance, biofilm formation, secondary metabolism, and microbial mutations. Accordingly, recent research indicates the impacts of simulated microgravity on human physiology and bacterial characteristics. This paper has reviewed the aspects of microgravity on changes in microbiota, including virulence, antibiotic resistance, and gene expression. Microgravity can undermine humans and makes influence bacterial pathogenicity. The review of papers shows that some microorganisms showed higher pathogenicity under microgravity conditions. Moreover, sulfamethoxazole had the highest resistance among Gram-positive microorganisms, and gentamicin had the highest resistance in Gram-negative bacteria. All antibiotics reviewed under microgravity conditions were robust in both groups of microorganisms compared to the gravity condition. Furthermore, some gene expression was altered in bacteria under microgravity conditions compared to Earth conditions (standard bacterial growth conditions). Changes in microbial behavior under microgravity directly influence astronauts' health conditions, and a detailed analysis of known facts can provide essential information for the selection of appropriate probiotics for these specific cases during the missions and after the recovery processes. Moreover, the study of microorganisms changes in the absence of gravity will help to understand the mechanisms of causing diseases on Earth and may be applied in clinical practice.

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1. INTRODUCTION

The environment beyond the Earth's gravity and a magnetic field is exceedingly severe and imposes many stressors on astronauts, including microgravity, cosmic rays, high vacuum, disruption of the circadian cycle, alterations in body fluids, isolation, stress, and changes in sleep patterns. Understanding

the possible effects of the alterations and generated problems for astronauts' body vital systems is crucial to maintaining their health during flight and after return to Earth. The knowledge built from previous flight research confirmed that astronauts suffer from a wide range of problems, including cardiovascular disorders, loss of muscle and bone tissue, immune depression, gastrointestinal disorder, respiratory distress, and some infections after long-duration space missions [1]. Moreover, astronauts generally need a long-term recovery after space missions under a health professional's control.

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All humans have many symbiotic bacteria on their skin and inside their bodies, generally refer as host microbiota. The microbiota plays an essential role in maintaining human health. It is actively involved in protection against a range of disorders such as systemic metabolic illnesses, including type 2 diabetes and obesity [2], inflammatory bowel diseases [3], allergic illness [4], autism spectrum disorders [5, 6], and schizophrenia and atopic eczema [7]. Other studies have revealed valuable properties against cancer [8-10]. However, in case of disruption or lack of commensal microbiota, the pathogens can cause systemic infections [11].

By general practice, astronauts are supposed to be in good health, without any chronic diseases, and usually are subject to long-term medical observations before being sent on missions. When the astronaut leaves the Earth's atmosphere and enters the stressful environment of space, all their microorganisms are also affected by the new conditions of that unique environment [12]. Research has approved that to cope with extreme space conditions; microorganisms inevitably undergo some changes. It is worthwhile to explore how microorganisms react to these conditions and what kind of changes occur in the metabolisms, especially under microgravity. It has been reported that exposure to space and ground-simulated microgravity (SMG) has great effects on microorganisms, such as changes in physiology, morphology, gene expression, growth rate, secondary metabolism production, biofilm formation, virulence, pathogenicity, resistance to environmental stresses including antibiotics [13-15]. Indeed, a well-known fact is that the immune system of humans (and other animals) is jeopardized in space [16], and thus can be even more relevant to the alterations in the behavior of the human microbiota. *Escherichia* (*E.*) *coli* as a Gram-negative (G^-) bacteria, has been studied [17] as the best-known microorganism at biochemical, molecular-genetic, and physiological stages. Although most *E. coli* strains are not pathogenic, some serotypes may lead to significant poisoning and infections for humans, including astronauts [18]. The standard treatment for such conditions is gentamicin. Most of the studies on *E. coli* behavior have been conducted in Earth-orbiting vehicles [14] in a state of simulated microgravity (SMG) [19], in which pressurized modules and life support systems may preserve the microbes from severe stresses. In several studies, microgravity and radiation have also been investigated [20, 21], while the focus in a few reports is mainly based on microgravity alone [22]. In this review, we are gathering the microgravity effect on physiology (*i.e.*, stress resistance and antibiotic resistance and the virulence of both G^- and Gram-positive (G^+) microorganisms; and specific gene expression. More investigations should be conducted to show whether these responses are due to an indirect and or direct effect of microgravity.

2. ALTERATION IN HUMAN MICROBIOTA UNDER SPACE FLIGHT CONDITIONS

As mentioned above, space traveling is accompanied by several risks [23], including psychological tension [24], strong abnormal radiation, and decreased gravity. Millions of years of evolutionary processes have adapted all kinds of life forms on Earth to the specificity of environmental conditions. Therefore, microgravity is an extraordinary experience

for the inhabitants of the Earth. In these critical conditions, shear stress and turbulence are reduced due to the elimination of convective currents, and as a consequence microbial physiology may be affected [25, 26]. Previous studies have shown that mobility is the most influential factor in the behavior of microorganisms in microgravity conditions [27]. In a way, a decrease in lag phase and an increase in cellular number have been observed only in immotile bacteria in microgravity conditions compared to normal gravity environment [28, 29]. Microgravity is considered a static condition in that the culture medium is not disturbed. In such a situation, the motile bacteria can have the advantage since they can mix the microenvironment with their flagella, and mass transfer can be performed well. Therefore, the presence or absence of gravity will not make a difference in their number. However, due to the removal of sedimentation, a process consequence of gravity, distribution of available food, air bubbles, and wastes, immotile bacteria have changed in microgravity, so increases in their number may be observed [30]. Moreover, some recent model studies also indicate the impact of microgravity on microorganisms and the human body [31].

2.1. Microbial Life in Space (Growth, Rate and Stability)

Some microorganisms can perceive mechanical and physical stimuli in their living environment, and exposure to some of these stimuli has effects on them [32]. According to the hypothesis of microbial life in space [33], a few extremotolerant and extremophilic microbes can survive in the stressed environmental condition of extra high or low temperatures and pressures [34-36], low pH, ethanol, oxygen levels, acid-resistant, osmotic pressure gradients [37], high radiation, osmosis pressure, and media limitations of nutrients [38]. It was shown that *Salmonella* (*S.*) *typhimurium* in microgravity conditions possesses enhanced resistance to osmotic acid and thermal stresses and survivability in macrophages compared to the normal gravity condition [39]. Moreover, it seems that the enhanced pathogenicity and stress resistance recognized in some *Salmonella* strains have relied on the sigma factor (RpoS) [40]. Phages, transposons, and plasmids play an essential role in bacterial adaptation and evolution. Also, many researchers have studied the bacteria's growth responses to microgravity conditions [41-44].

Literature review shows that in microgravity conditions, acid and heat stress resistance has been changed (increased or decreased) in bacteria compared to a normal gravity environment. Some of these results have summarized in Fig. (1). However, in some bacterial species, such as *S. typhimurium* [41, 43, 44] or *Enterobacter* (*E.*) *cloacae* [43], contradictory observation has been reported which is due to differences in bacterial species and/or culture conditions [42]. In most cases, alterations in the expression of some related genes are mainly responsible for such differences [42-47].

Among all types of bacterial species, the space condition studies on bacteria are majorly oriented toward *E. coli* [48], *Bacillus* (*B.*) *subtilis* [49] *Pseudomonas* (*P.*) *aeruginosa* [50], *Serratia* (*S.*) *marcescens* [51], *Enterococcus* (*E.*) *faecalis* [52], *Rhodospirillum* (*R.*) *rubrum* [21], *S. typhimurium* [53], *P. aeruginosa* [54], *Staphylococcus* spp. [55], *Strepto-*

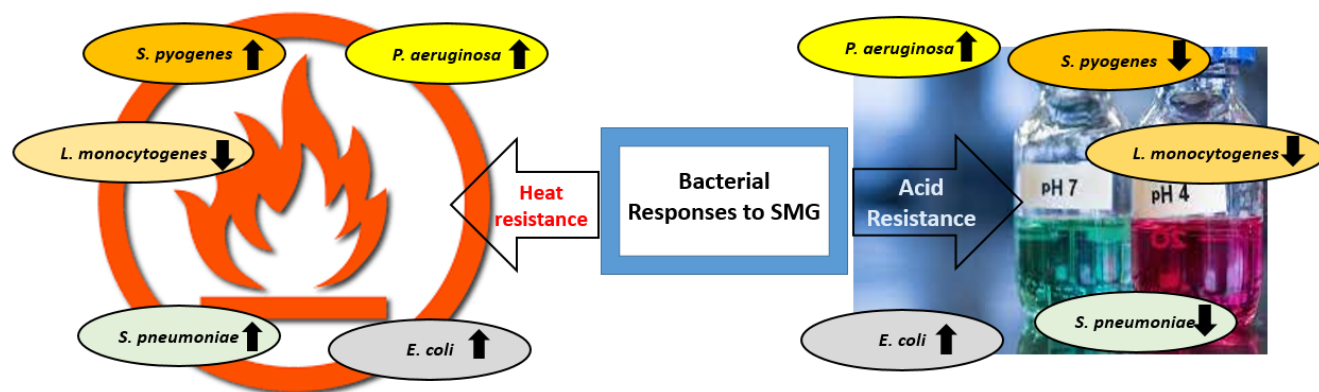


Fig. (1). Some bacterial resistance changes to acid and heat stresses in SMG conditions compare to normal gravity [1-5]. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

coccus spp. [56], and *Streptomyces* spp. [57], since most of them are well-known human and other animal pathogens, their behavior is directly associated with astronauts' health. Besides the bacterial species, *Saccharomyces* (*S.*) *cerevisiae* [58] and *Candida* (*C.*) *albicans* [59] have also been studied regarding their behavior in conditions simulating low gravity.

Generally, it can be concluded from all previously reported studies that microgravity conditions lead to decreased lag phase and enhanced exponential growth. Therefore, the final population density of the cell was approximately doubled. Also, an increased exponential growth rate in space was observed. On the other hand, studies indicate that microgravity does not influence the exponential growth rate of *E. coli*, which depends on the motility and ingredients of the medium and the bacterial strain and species [60]. Besides, microgravity facilitates growth in the measure of nonmotile cells in the submerged medium. Exception of motile bacteria is *E. coli*, *Salmonella* spp., *Pseudomonas* spp., and *Bacillus* spp. An enhanced growth rate and biomass yield of the nonmotile *Micrococcus* (*M.*) *luteus* has also been reported. In addition to assess the effect of reduced gravity on bacterial properties, human physiology has been studied in these conditions. The corresponding impacts on the human body are only sparsely studied; thus, the results indicate that the human's immune response is compromised when they are in the microgravity of space. The suppressed immune system corresponds to a high risk of infectious illnesses and antimicrobial resistance, which is a critical risk for astronauts. Thus, considering accelerated cell growth and increased biomass from one side and suppressed immune system from the other is an alarming fact directly associated with the health of astronauts.

2.2. Body Health and Gut Microbiota of Astronauts

Space exploration has influenced the human body positively and negatively [61]. Besides the different detected adverse biological influences on the human due to an extended space journey such as loss of muscle mass, decrease in bone density, kidney stones, impaired cardiac performance, lack of balance, varying pharmacokinetics and phar-

macodynamics, diminished vision, etc. [62], several studies have been indicated the anticancer impacts of the microgravity [63].

Investigations have shown that microgravity has a profound effect on the cytoskeleton [64, 65], extracellular matrix [65], and processes such as apoptosis [66] and gene expression [64, 65] on thyroid cancer cells in a way that has reduced malignancy in this cell line. However, apoptosis induction by microgravity has also been reported in some other cell lines, such as lymphocytes [67] and breast cancer cell lines [67].

Special space conditions, such as microgravity, radiation, and a high iron diet, may have adverse effects on astronaut health due to changes in intestinal microbiome balance, especially in long-term missions [68, 69]. Such microbial changes have been reported as an alteration in their composition and function [70]. Skin microbiota balance disruption may also cause skin diseases and infections [71].

However, in case of disruption or lack of commensal microbiota, the pathogens can cause systemic infections [11]. Therefore, it is critical to sustaining the balance of microbiota to impede infections in space stations for astronauts. However, little information is accessible indicating the aspects of microgravity on the function of gut microbiota, except for a few types of research demonstrating the major reduction of *Lactobacillus* (reclassified recently to different genera) and *Bifidobacterium*, and increase in several virulent bacteria such as *E. coli*, *Clostridium*, *Fusobacterium* (*F.*) *nucleatum* and *P. aeruginosa* and *Proteus* spp., due to the diverse ecological factors that may take place in space such as radiation, stress, and diet under microgravity conditions [1]. Accordingly, astronauts' immune systems and immunological responses are drastically suppressed [72]. It may be acknowledged that by several reasonable arguments, astronauts live and work in almost sterile environmental conditions, and this as well influences the functionality of the immune system. However, research considering the effects of microgravity on bacterial cells and the human body is bringing much concern since corresponding data will enhance awareness and utilization of biotechnology in space.

The SMG is significantly acceptable and available to most scientists for the sake of reduced expenses, without any additional requirement of space journey and highly responsive facilities. Hence, a group of microgravity simulator apparatuses was developed [73].

3. Microgravity Simulators

Due to the limitations associated with performing experiments in real space microgravity, such as astronaut health concerns, exorbitant costs, special equipment requirements, and limited reproducibility, simulator devices are used for preliminary bio-space experiments, especially microgravity simulators [13, 74]. Microgravity simulators are devices that simulate real microgravity by changing the gravity vector steadily so that the gravity consequent is close to zero [14, 75].

Lack of fluid convection in microgravity has led to the introduction of space as a static environment, with low hydrodynamic shear stress and no mass and heat transfer [76]. In microgravity conditions created by ground simulators, gravitational forces are very small (10^{-3} - 10^{-6} g) and not entirely the same as real microgravity (10^{-6} g). Therefore, in most studies, the simulated microgravity phrase is considered "approximately" equivalent to real microgravity [77, 78].

Some of these microgravity simulator devices are clinostats [79], random positioning machines [80], rotating-wall vessels [81], and diamagnetic levitation apparatus [82]. Exposure requirement is the selection criteria [13]. Clinostats and random positioning machines have been more suitable for studying microgravity effects on higher organisms [83], while rotating-wall vessels have been used on different cell types, from microorganisms to higher organisms [13]. Unlike the mentioned devices in which rotation is the principle of their microgravity simulation, at the sample site on the diamagnetic device, the magnetic force and the microgravity approximately neutralize each other so that gravitational force went near zero [75]. This simulator has been employed on many kinds of organisms, from microorganisms to higher organisms as well [13].

Recently, several specific research projects have demonstrated different aspects of microgravity on microbial pathogenesis, antibiotic resistance, and gene expression focused on rotational apparatus in SMG [27].

4. EFFECT OF MICROGRAVITY ON MICROBIAL VIRULENCE OF PATHOGENIC MICROORGANISMS

Recent studies indicate that microorganisms would become a more dangerous and present health risk to astronauts under microgravity conditions due to enhanced availability of nutrients, more quick bacterial cell division decreased lag phase, enhanced population growth, increased secondary metabolite production, less fluid shear forces, Brownian motion, biofilm formation, genomic changes (including the fact of increased levels of radiations in the orbital stations and consequence of stimulated mutation processes), increased catecholamine expression, and resilience concerning the acidic stress [84]. Microorganisms are present and survive

onboard the International Space Station (ISS), which may have been introduced with new cargoes and space crew or stationed there since the resumption [85]. Besides, studies have shown that the ability of astronauts' immune systems declined during long space exploration [86]. Therefore, it may be possible for opportunistic pathogenic bacteria to infect crew members [22, 87-90]. The effects of microgravity on bacterial virulence have summarized in Table 1.

Nickerson *et al.* (2000), Belay *et al.* (2002), and Aviles *et al.* (2003) revealed that *in vivo* conditions, microgravity, could increase *Salmonella* (*S.*) *enterica* serovar Typhimurium, *Klebsiella* (*K.*) *pneumonia*, and *P. aeruginosa* virulence. In the vaccination of mice under SMG, initially, a significant increase rate of a week post-infection was observed. Then, one week and a half post-infection, the variations were more enounced, with only twenty percent of mice staying alive after vaccination compared to the sixty percent of those infected with control *Salmonella*. Chopra *et al.* (2006) indicated that the mouse mortality rate of microgravity-exposed *S. enterica* serovar Typhimurium increased from sixty percent at a half week post-infection to a hundred percent lethality applying a 1×10^7 CFU/ml virulent dose, which was conceivably hyper-virulent. Other studies have shown an increase in virulence under microgravity *in vitro* conditions, including *Salmonella* spp. [91], *S. marcescens* [51], *E. coli* [27, 92, 93], *B. subtilis* and *Staphylococcus* (*St.*) *aureus* [94], *P. aeruginosa* [92], and production of a larger measure of exopolymers substances (EPS) by *M. luteus* [88]. *P. aeruginosa* constitutes the most biofilm as observed for *E. coli* [90]. Accordingly, *P. aeruginosa* is conceivably hyper-virulent [50] because biofilm formation is a more significant factor of microbial virulence [42]. On the other hand, the study by Hammond *et al.* (2016) demonstrated that *Streptococcus* spp., *Enterococcus*, and *Lactobacillus* (*Lb.*) *acidophilus* are not virulent and have the same behavior as those on the Earth. Vesper *et al.*, (2008) indicated that the fungi include a huge variety of conceivably pathogenic molds, namely, *Aspergillus* (*A.*) *niger* and *Aspergillus* (*A.*) *flavus*, and toxigenic molds, namely, *Penicillium* (*P.*) *brevicompactum* and *Penicillium* (*P.*) *chrysogenum* were virulent, while *Aspergillus* (*A.*) *terreus*, *Aspergillus* (*A.*) *fumigatus*, *C. albicans*, or *Fusarium* (*F.*) *solani* were not virulent in SMG condition.

According to information provided in Table 1, among the studied microorganisms, the effect of pathogenicity under microgravity was mostly investigated on G⁻ bacteria, and most pathogenic microorganisms are also found among these bacteria. It can be mentioned that *S. enterica* serovar Typhimurium was only studied on mice *in vivo* conditions in a hyper-virulent manner. For instance, *Salmonella* spp. is perfectly sensible to alternations in osmolality, pH, growth phase, and stress [95]. Besides, *S. enterica* serovar Typhimurium is the primary reason for common infections in the human body, particularly in people with immunodeficiency. Therefore, it can be mentioned that astronauts have a more debilitated immune system than individuals during their space journey. Thus, a comprehensive investigation of these microorganisms under SMG conditions is very important. Moreover, the G⁻ bacterium *P. aeruginosa* has become hyper-virulent under microgravity compared to other microorganisms [45, 50]. Therefore, due to the enhancement in

Table 1. Bacterial virulence changes in response to microgravity.

Microorganisms		Culture Condition				Bacterial Responses to Microgravity		Evidence	Ref.
		Space		SMG					
		<i>in vivo</i>	<i>in vitro</i>	<i>in vivo</i>	<i>in vitro</i>	Increased Virulence	Decreased Virulence		
G ⁺	<i>Micrococcus luteus</i>		✓			✓		Grew faster and production of a larger measure of exopolymeric substances	[88]
	<i>Streptococcus</i> spp.		✓		✓		✓	Decreased the lethality of wild-type <i>Caenorhabditis elegans</i>	[56]
	<i>Enterococcus</i> spp.		✓		✓		✓	Decreased the lethality of wild-type <i>Caenorhabditis elegans</i>	[56]
	<i>Lactobacillus acidophilus</i>				✓	--	--	No significant differences between microgravity and grown control	[89]
	<i>Listeria monocytogenes</i>				✓		✓	Diminished transcription for virulence-related genes	[42]
G ⁻	<i>Escherichia coli</i>		✓		✓	✓		Biofilm thickening and increasing resistance to the general stress (salt, ethanol, and some antibiotics)	[90]
	Salmonella enterica serovar Typhimurium				✓	✓		Increased numbers of bacteria from the murine liver and spleen following oral infection, LD ₅₀ value 5.2 times lowered, and expression of around 100 genes was altered	[53]
	Salmonella spp.				✓	✓		LD ₅₀ was lowered, and mice survival and the time of death post-infection were decreased	[91]
	Pseudomonas aeruginosa		✓		✓	✓		Increased Biofilm thickening and biomass of bacteria, Increased expression of virulence-related genes	[45, 50]
	Serratia marcescens					✓	✓	Increased lethality against <i>Drosophila melanogaster</i> after growth in ISS	[51]
	Klebsiella pneumonia	✓					✓	Decreased the lethality of wild-type <i>Caenorhabditis elegans</i>	[56]

G⁺: gram-positive, G⁻: gram-negative.

antibiotic resistance and the consequent growth in pathogenicity in space, it can be noted that *P. aeruginosa* indicates high pathogenicity under microgravity conditions compared to the normal gravity condition.

5. EFFECT OF SMG ON MICROBIAL ANTIBIOTIC RESISTANCE

One of the main concerns in the microgravity condition is that microbial antibiotic resistance could be enhanced [1, 42, 44, 96] due to different reasons, such as intricate interactions between main stress factors (space radiation and microgravity) [97], a decrease of the colloidal EPS production [98], biofilm formation [88], reductions in cell size about 73%, and increase in cell numbers because of thickening membranes of the amplified specimens [99]. Another reason can be associated with a lack of the actual effects provided by gravity, *i.e.*, sedimentation and buoyancy, that can change

the way bacteria take in nutrients or drugs intended to attack them [100]. One of the main differences between *E. coli* and other microorganisms is that *E. coli* is more implicated in forming clustered structures, thus, preventing outer cells during the antibiotic injection to keep the inner cells well [101]. Hence, it can be mentioned that decreasing antibiotic resistance is an important factor in the space environment [58].

Many studies have investigated the effects of microgravity on antibiotic resistance. Mora *et al.*, (2016) studied the potential of nineteen bacteria samples to survive a heat shock and antibiotic resistance *in vitro* conditions. It can be noted that only *Micrococcus (M.) yunnanensis* showed no resistance over the entire investigated antibiotics. In contrast, *Paenibacillus (Pa.) campinasensis* showed the most resistance, indicating resistance against eight of the seventeen investigated antibiotics [102]. In 19-month research, the percentage of antibiotic

Table 2. Effect of antibiotic resistance G⁺ and G⁻ bacteria under SMG *in vitro* condition.

Microorganisms	Antibiotic Resistance to	More Resistant to	Ref.
G ⁺	<i>Streptococcus pyogenes</i>	Ampicillin, Rifampicin	[47]
	<i>Lactobacillus acidophilus</i>	Sulfur gentamicin	[104]
	<i>Staphylococcus aureus</i>	Sulfamethoxazole, ampicillin, and erythromycin	[16,105]
	<i>Staphylococcus epidermidis</i>	Rifampicin, sulfamethoxazole, and erythromycin	[76]
	<i>Staphylococcus hominis</i>	Sulfamethoxazole	[16]
	<i>Streptococcus pneumoniae</i>	Several antibiotics	
	<i>Enterococcus faecalis</i>	Kanamycin, gentamicin, sulfamethoxazole, tetracycline, doxycycline, clindamycin, chloramphenicol, meropenem	
	<i>Bacillus</i> spp., <i>Propionibacterium</i> spp., <i>Corynebacterium</i> spp.	Tetracycline, clindamycin	[103]
	<i>Bacillus anthracis</i>	Cefotaxime, ceftriaxone	[110, 111]
G ⁻	<i>Escherichia coli</i>	Gentamicin, colistin, kanamycin, ciprofloxacin, cefotaxime, chloramphenicol, cefalotin, cefuroxime, cefuroxime axetil, ceftazidime, tetracycline, penicillin G, ampicillin, nalidixic acid	[15, 17, 106]
	<i>Serratia marcescens</i>	Fosfomycin	[111]
	<i>Enterobacter bugandensis</i>	Cefazolin, ceftazidime, oxacillin, penicillin, rifampin, ciprofloxacin, fosfomycin, amikacin	[107]
	<i>Salmonella enteritidis</i>	Amikacin	[19]
	<i>Haloferax mediterranei</i>	Bacitracin, erythromycin, rifampicin	[108]

resistance among isolated bacteria was studied [103]. The bacterial species demonstrated higher sensitivity to the antibiotics, while only a few species indicated enhanced resistance. Besides, sulfamethoxazole and erythromycin showed the most resistance.

Moreover, ampicillin resistance was repeatedly identified as harmful, often associated with sulfamethoxazole and erythromycin resistance. In a study by Be *et al.* (2017), the microorganisms were resistant to β -lactams, fluoroquinolones, aminoglycosides, clindamycin, lincosamides, tetracycline, and streptomycin. Moreover, Schiwon *et al.* (2013) indicated that enterococci and staphylococci form considerable biofilm.

Despite their thicker peptidoglycan layer on the Earth, the G⁺ bacteria are more susceptible to particular cell walls targeting antibiotics than G⁻ bacteria because of the outer membrane absence. However, it is still a challenging issue if G⁺ bacteria or G⁻ bacteria are resistant to an SMG medium. The recent literature review on the effect of antibiotic re-

sistance of G⁺ and G⁻ microorganisms on microgravity is described, and a summary is provided in Table 2. Kalpana *et al.* (2016) showed that *St. pyogenes* in microgravity are more resistant to rifampicin and ampicillin compared to the normal gravity cultures. In Shao *et al.* (2017), *Lb. acidophilus* was more resistant to sulfur gentamicin compared to other investigated antibiotics in the SMG condition. Accordingly, antibiotic resistance can be illustrated as sulfur gentamicin > cefalexin > chloramphenicol > sodium penicillin.

Results presented by Vaishampayan and Grohmann (2019) demonstrated that *Bacillus* spp., *Staphylococcus* spp., *Propionibacterium* spp., *Enterobacteriaceae*, and *Corynebacterium* spp. are resistant to antibiotics. The resistance of *E. faecalis* and *Staphylococcus* spp. was due to transfer genes. The result presented by Schiwon *et al.* (2013) showed that two-thirds of the bacteria isolates are resistant to at least antibiotics, in which more than 80% of the bacteria formed biofilms. Besides, *St. aureus* showed two times more resistance compared to oxacillin, chloramphenicol, and

Table 3. Comparison of antibiotic resistance of G⁺ and G⁻ microorganisms in normal and Microgravity conditions

Target Group	Antibiotic Substance	Type	Resistance in Gravity	Resistance in Microgravity	Mechanism of Action	Ref.
G ⁺	Ampicillin	β -Lactam antibiotic penicillin	✗	✓ (SMG)	Impedes cell wall synthesis	[47]
				✓ (Real Microgravity)	Impedes cell wall synthesis	[102]
	Clindamycin	Lincosamide	✗	✓ (Real Microgravity)	Inhibits protein synthesis	[102]
	Tetracycline	polyketide antibiotic	✓		Interfering with protein synthesis	
	erythromycin	Macrolide	✓	✓ (Real Microgravity)	Inhibits protein synthesis	[102, 16]
	sulfamethoxazole	Dihydrofolate reductase inhibitor and sulfonamide	✗	✓ (Real Microgravity)	Inhibits tetrahydrofolate synthesis	
G ⁻	Rifampicin	medications called antimycobacterial	✓	✓ (Real microgravity)	Inhibiting bacterial DNA-dependent	[112]
	Gentamicin	Aminoglycoside	✗	✓ (Real Microgravity)	Impedes protein synthesis	[102]
	Chloramphenicol	chloramphenicol	✗	✓ (Real Microgravity)	Inhibiting peptide bond formation and subsequent protein synthesis	[102, 16]
	Ciprofloxacin	Fluoroquinolone	✓	✓ (Real Microgravity)	Inhibits bacterial DNA gyrase	[102]
				✓ (SMG)		[93]
	Penicillin G	β -Lactam antibiotic penicillin	✗	✓ (Real Microgravity)	Impedes cell wall synthesis	[102]
	Cefotaxime	β -Lactam antibiotic; cephalosporin	✓	✓ (Real Microgravity)	Impedes cell wall synthesis	[102]
				✓ (SMG)		[17]
	cefalotin	cephalosporin	✓	✓ (SMG)	Inhibition of cell wall synthesis	[17]
	Colistin	Polypeptide antibiotic	✓	✓ (Real Microgravity)	Attacks cell membrane	[102, 113]
	kanamycin	aminoglycoside	✓	✓ (Real microgravity)	Changes in ribosome structure	[109]

✓: resistance, and ✗: no resistance.

erythromycin in SMG conditions [104-109]. A similar study indicated that *St. aureus* is resistant to several antibiotics, such as penicillin, rifampin, and erythromycin [105]. In results presented in reference to Sobisch *et al.* (2019), *S. aureus* showed resistance of 89% to erythromycin and ampicillin, 33% to kanamycin, and 12% to clindamycin and tetracycline microgravity. Moreover, in microgravity conditions *Staphylococcus (St.) hominis* showed resistance of 95% to sulfamethoxazole, 82% to erythromycin, 62% to ampicillin, and 24% to clindamycin, 28% to tetracycline, 10% to kanamycin, 7% to meropenem, and 4% to doxycycline and tigecycline. Furthermore, *Staphylococcus (St.) epidermidis* showed

resistance of 92% to erythromycin, 24% to ampicillin, 9% to doxycycline, and the most resistance to sulfamethoxazole in microgravity. The *E. faecalis* in microgravity showed resistance of 65% to doxycycline and tetracycline, 65% to clindamycin, 34% to chloramphenicol, and 34% to meropenem, and most resistance to kanamycin, sulfamethoxazole, and gentamicin. Furthermore, it was concluded by Tixador *et al.* (1985) that *E. coli* is 60% resistant to gentamicin in SMG due to the presence of σ_s , inscribed by the *rpoS* gene. Moreover, the antibiotics colistin and kanamycin revealed over fourfold enhancement against *E. coli* than the Earth controls. In the study of Doğanay and Aydin (1991) and Rigsby *et al.*

Table 4. Gene expression and function of some stress response genes under SMG *in vitro* condition.

Microorganisms		Gene function	Expression of stress response genes	Ref.
G ⁺	<i>Staphylococcus epidermidis</i>	Mutation of the <i>rpoB</i> gene, encrypting the rifampicin resistance protein	Expression of the <i>rpoB</i> gene	[112]
	<i>Lactobacillus reuteri</i>	Increased production of reuterin and higher tolerance to the gastrointestinal passage	tolerance stress Varying	[60]
	<i>Staphylococcus warneri</i>	Increasing the resistance and flexibility of bacteria	Unregulated phosphotransferase system	[121]
	<i>Staphylococcus aureus</i>	Decrease in expression of virulence	Decreased approximately two-fold isolate RF6, the expression of <i>hla</i> and genes <i>saeR</i> / <i>saeS</i>	[122]
	Lactobacillus acidophilus	Reduction in cholesterol	Genes such as <i>abcb11</i> , <i>hmgcr</i> , <i>cyp7a1</i> , and <i>ldlr</i> in the HepG2 cell line	[104]
G ⁻	<i>Escherichia coli</i>	Enhanced biofilm	3-4 fold increased expression of <i>acrA</i> , <i>arcB</i> , and <i>tolC</i> genes	[93]
		Enhancing viable	A fold alternation of 1.5 <i>ydcQ</i> , <i>yniB</i> , <i>ygaQ</i> , <i>yffD</i> <i>ybdJ</i> , <i>ynfA</i> , and <i>hokE</i> genes	[123]
		Enhanced growth	Enhancement in transcripts for DNA replication	
		Decrease stress resistance	Whit increases RpoS protein	
		Enhance in response to acid stress	Two-fold increased the expression of <i>hdeA</i> , <i>hdeB</i> , and <i>rpoS</i>	[124]
	<i>Pseudomonas aeruginosa</i>	Enhanced growth	Regulation of 27 proteins and 168 genes	[54]
	<i>Serratia marcescens</i>	Heat shock protein	HslU 2.31	[51]
	<i>Aliivibrio fischeri</i>	Minimal effect on symbiont gene requirement	RodA was more depleted, change one gene (<i>rodA</i>)	[125]
	<i>Salmonella typhimurium</i>	Enhanced the virulence	change proteins and genes <i>spi-1</i> and <i>spi-2</i>	[53]
		LPS levels change	Decreased genes expression of <i>rfb</i>	
		Ability to deal with macrophages and enhanced resistance to osmotic, thermal, and acid stresses	Expressed 167 genes differentially 167 transcripts and 73 proteins (69 up-regulated and 98 down-regulated)	
		Enhanced the virulence	Reduced genes of <i>spi-1</i> , <i>spi-2</i> , and changed proteins	
		Change phenotypes	Change RpoS protein	[41, 44]
		Thermal stresses	Change RpoS protein	
		Enhance growth kinetics	Reducing of expression of <i>hfq</i>	
		Effects on metabolism	Changing of 65 genes expression in the <i>hfq</i>	
		Enhanced the sensitivity to oxidative stress	Change of expression of f <i>hfq</i>	
		Enhanced metabolic activity whit minimal media	Many genes exhibited decreased expression	
		Biofilm production	Gene <i>ydcI</i>	[126]
		Acid resistance	<i>rpoS</i> gene-encoded product σs	
		Reduced peroxide	Altering system soxRS, adjustment of <i>oxyR</i> and <i>soxS</i> gene patterns, recognizing two systems of toxin-antitoxin genes, signaling molecule indole	

Yeast	Change of stress resistance	<i>yil052c</i> , <i>yst2</i> , and <i>ssa4</i> with transcription adjustment of <i>msn4p</i> and <i>sfp1p</i>	[127]
	Overexpressing grown in microgravity	Overexpressing one specific <i>flo</i> gene	[128]
	Varying cellular functions	overexpressed genes in some cases, more than 2000	
	catabolize glucose into organic acids	Up-regulated genes <i>poxb</i>	
	Resistance to extreme acid conditions	<i>gadABC</i> family genes	
Saccharomyces cerevisiae	Shortened lag phase (90 min)	Expression changes	[127]
	Aggregates formation along with a rare adjustment of <i>egt2</i> , <i>dse1</i> , and <i>dse2</i>	36% reduction of <i>ssa4</i> , 44% reduction of <i>yil052c</i> , 0.07% reduction of <i>yst-2</i>	
		Gene expression of <i>ssa4</i>	
		<i>yil052c</i> is dependent on <i>sfp1</i>	
		Gene expression changes of the <i>yil052c</i> , <i>yst2</i> , and <i>ssa4</i> genes	
Candida albicans	Change morphology	2-fold increase in <i>hwp1</i> , a 25-fold decrease in <i>ywp1</i>	[59]

(2005), a large number of resistant to antibiotic strains were detected during the investigation of the cephalosporines cefotaxime, and ceftriaxone isolates as 16 out of 19 cefotaxime and 14 out of 19 ceftriaxone. Besides, *Bacillus* showed the most resistance among G^+ isolates. Generally, the *Bacillus* species reported party resistance in contact with ceftriaxone and cefotaxime, as demonstrated previously for *Bacillus (B.) anthracis*. Furthermore, some studies [110, 111] indicated that *S. enterica* is resistant to ciprofloxacin due to a mutation in the *gyrA* gene, and *S. marcescens* is also resistant to fosfomycin as a consequence of the *FosA* protein-coding gene presence. Zhang *et al.* (2019) demonstrated that the resistance of amikacin was due to the existing *oppA* gene and oligopeptide transporter protein *OppA* in *Salmonella (S.) enteritidis*.

According to data presented in Table 2, among 18 types of antibiotics studied on nine strains of G^+ microorganisms, sulfamethoxazole, erythromycin, and ampicillin indicated the highest resistance, respectively. Gentamicin also showed the most resistant over 14 types of antibiotics studied on four strains of G^- bacteria.

A comparison of antibiotic resistance of G^+ and G^- bacteria in normal and microgravity conditions is demonstrated in Table 3. As shown in Table 3, all reviewed groups of microorganisms were resistant to antibiotics in different percentages under microgravity conditions. Among these antibiotics, erythromycin and tetracycline were not affecting G^+ microorganisms in terms of gravity, and ciprofloxacin, colistin, cefotaxime, and cefalotin were not affecting tested G^- microorganisms. Resistance to kanamycin and rifampicin have also been reviewed in the gravity condition on both groups of microorganisms in which G^+ microorganisms were only resistant to rifampicin and kanamycin was not affecting the growth of G^- ones.

6. EFFECT OF SMG ON MICROBIAL GENE EXPRESSION

Space journey and the corresponding stress condition could influence the gene expression or even the entire chromosome of cells [112-114]. Microorganisms undergo such changes to adapt and survive in the space environment. However, gene expression studies may help predict microbial behavior after exposure to stressful situations, especially microgravity, and the results are useful for medical bioengineering applications [115]. The first microbial microarray study under microgravity was represented in 2002 [44], and the whole genome of *S. enterica* serovar Typhimurium was elucidated in response to SMG. The results showed that SMG had changed the expression of the 163 genes compared to normal gravity. Such alterations occurred in functionally diverse groups, including virulence factors, transcriptional regulators, iron-utilization enzymes, lipopolysaccharide biosynthetic enzymes, and proteins of unknown function [44]. Moreover, similar studies followed in microorganisms' whole genome gene expression alteration in both space flight [116, 117] and ground base microgravity [23, 118, 119]. Recognizing such changes will help to overcome microgravity conditions and maintain the health of astronauts.

De Boever *et al.* (2007) revealed the effective gene relocation in *B. thuringiensis* among G^+ microorganisms in space journeys in comparison with the Earth control, while *E. coli* and *Cupriavidus (C.) metallidurans* did not show gene relocation among G^- microorganisms [120]. The extension and enhancement of bacterial pathogen and antibiotic resistance in the space journey can be associated with practical gene transfer actions [17]. In a work performed by Aunins *et al.* (2018), 1,980 out of 5,406 examined genes were identified as adaptable among investigations on the Earth and space. Except for almost 27 genes out of the entire examined, genes

showed highly alterable in SMG investigations. Burgos *et al.* (2020) identified toxin and antitoxin genes and the signaling molecule indole, which plays a significant function in the variations of microorganisms in SMG for particular stress integration. The gene expression and function of stress response genes studied in the literature review on microgravity are summarized in Table 4.

According to results presented in Table 4, the gene expression is generally increased in some bacteria genes under microgravity conditions compared to Earth control, which causes stress resistance. In this regard, we can make probiotics and generally recognized as safe microorganisms, resistant to stresses by manipulating four effective and repetitive genes, namely, sigma factor *rpoS* [121-129], small RNA binding protein *hfq* [130], *rodA* [131], and *flo* [132-135] genes in bacteria (the function of these genes were mentioned on Table 4). This is because the above-mentioned genes play an essential role in resisting stressful conditions.

CONCLUSION

To ensure the health of astronauts and minimize their requirements in space, it is necessary to design an efficient life support system. In most studies, changes in the astronaut's immune system in space conditions have been investigated. On the contrary, alteration of astronaut microbiota or those microorganisms that have been transported to space by equipment have received less attention. Microbiota is an inseparable part of astronaut bodies, and their count and composition may be affected by stressful conditions in space exploration, especially microgravity. These changes have been reported in the case of the composition and function of the microbiota under both simulated microgravity and real microgravity after 15 -35 days in space.

Considering that all living organisms on Earth have been created and evolved in the presence of gravity force, the investigation of the alterations that occurred in the absence of gravity will aid scientists in understanding the biological reaction mechanisms on Earth. Furthermore, this information may be applicable to clinical processes. Nowadays, due to long-time spaceflights, studying the correlation between humans and microorganisms in space conditions is significantly important to maintain the health of astronauts during the trip and after returning to earth. Since microbiota can be affected by the space environment, antibiotic resistance and virulence may be increased. This issue may be related to practical changes in gene expression. The mentioned alternations in microorganisms may be recognized as a potential threat to astronauts. Accordingly, the existing microbiota in space can be known as hazardous. Studies reviewed in this paper have demonstrated alternations in gene expression, biofilm formation, pathogenicity, and microbial interplay with their host and other investigated microorganisms. In this issue, increased pathogenicity of *S. enterica serovar typhimurium* (due to increased numbers of bacteria from the murine liver and spleen following oral infection) lowered LD₅₀ value 5.2 times, and altered expression of around 100 genes and *P. aeruginosa* (due to biofilm thickening increased, the biomass of bacteria enhancement and virulence-related genes overexpression compared to the other investi-

gated microorganisms) and increased resistance of sulfamethoxazole and gentamicin to other antibiotics is wondrous. As mentioned above, several space experiments and simulated microgravity tests have reported increased biofilm formation power for some bacteria compared to normal gravity on Earth. Since biofilm formation is a precursor to disease onset, in microgravity conditions, the binding power of bacteria and, as a result, the pathogenicity of microbiota bacteria will increase, and the health of space travelers will be threatened. Furthermore, based on the reviewed studies, it can be noted that the study of metabolites produced by the gut microbiota of astronauts under microgravity conditions is exceptionally vital. In addition, once the alternations in microbiota metabolites can be recognized as a result of microgravity conditions, it can be suggested appropriate solutions for prebiotics, probiotics, and postbiotics to deal with the pathological effects in astronauts.

AUTHOR CONTRIBUTIONS

Conceptualization, M.S. and K.K.D.; methodology, M.S.; software, S.D.T., A.D.T.; validation, M.S., Y.Y. and K.K.D.; formal analysis, S.D.T., and A.D.T.; investigation, S.M.A. and K.K.D.; resources, M.S., S.M.A., S.D.T., K.K.D.; data curation, M.S., S.M.A., K.K.D.; writing-original draft preparation, M.S., S.M.A., S.D.T., K.K.D.; writing review and editing, M.S., S.M.A., S.D.T., K.K.D.; visualization, A.D.T.; supervision, K.K.D.; project administration, M.S.; funding acquisition, K.K.D. All authors have read and agreed to the published version of the manuscript.

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CONFLICT OF INTEREST

The author declares no conflict of interest, financial or otherwise.

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